Team Control Number

13338

Problem Chosen

C

2022
HiMCM/MidMCM
Summary Sheet

Executive Summary

Theme parks are commonplace in our twenty-first century modern world. However, many of us are unsure of how to make the best use of our time at a theme park. There are those amongst us who prefer to be impulsive and try out everything rather than plan out activities beforehand. This can often lead to a poor and unfulfilling experience. To mitigate against this, MidMCM Travel Agency can plan out a range of activities for customers; this way customers will save time on researching the various activities yet still maximise their level of enjoyment. With this in mind, Ming, Ishmael, Karine, and Freya hired MidMCM team to plan their activities at Polygon Paradise Park.

Firstly, we planned the activity for each of the friends individually, based on their preferences and requirements. The Gantt chart is a powerful tool in project planning and management. In this project, we have made use of Excel to create the Gantt charts. The horizontal timeline includes the time of events being scheduled, ranging from 9am to 9pm. The other axis contains the name of the events that take place. To make it easier to understand, arrows are drawn to show the sequence of the tasks. The Gantt chart is also colour-coded: red cells represent tasks that must be finished within the specified timeframe while green cells are optional and can occur anywhere between a period. Before we design the Gantt chart, we first need to gather data. The data then can be extracted and represented onto a flow chart. Then, following the steps, and considering variables such as walking time from ride to ride; duration of each activity and their waiting time, the Gantt chart can be created.

Secondly, to gauge how much the group of friends enjoyed their day at the theme park, we developed a happiness mathematical model to calculate the "Happiness score". We estimated how much each friend would enjoy each activity and multiplied this by the duration of the activity. Whichever score is higher will be an indication of which schedule is the more successful one. Thus, we designed 9 scheduling cases in total, for example: one group of 4 friends, 2 groups of 2 friends, 2 friends combined in groups etc. According to the happiness mathematical model analysis, case 6 appears to be the best possible schedule; this schedule should be enjoyed by all friends, in this case, the group of friends only spilt up for 4 hrs of the day. The basics of the Gantt chart still apply, but in this case, there will be two charts with events happening simultaneously from both groups. However, we discovered that some the friends' requirements had to be compromised. Through this we concluded that separating the friends is crucial in maximising enjoyment for the whole group as the Happiness score is higher in the schedule with includes 4 hrs of division into subgroups than the entire 12 hrs duration.

Finally, we came up with the best route to maximise the enjoyment level of the group of friends, which is the optimal Gantt chart providing them with the highest enjoyment level. However, it is not possible for them to all their individual desires to be met by staying together form 9 am to 9pm, at some point during the day, they will have to split up in order to satisfy more requirements and maximise enjoyment. One detailed flow chart about the trip was also created for better visualization purposes.

With the increasing popularity of travel agencies providing this service, we recognise out level of accuracy will need to be very high and incorporates as many of the requirements and needs as possible of the individual within group; we need also to legislate for unexpected events.

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Chapter 1 Introduction

1.1 Problem interpretation

Our problem or 'challenge' is to plan a day of fun at Polygon Paradise Park for four friends. The park is open from 9am to 9pm. There are various activities in the amusement park, including rides of different types and thrill levels; a show with various performances throughout the day; food options, entertainments and other more basic facilities. We will have to meet the different requirements of the 4 friends to ensure maximum level of happiness. But no matter how perfectly we plan the activities, there will always be walking and waiting time. There is also a limitation that to maximise the enjoyment of the friends, it will be necessary to divide a 4 hrs period,

Preference of the 4 friends

-Ming cannot go on rides with extreme thrills or medium thrills after lunch. If not, he will throw up and severely lessen the enjoyment of himself and others. Ming also wants to win some prizes at the game's booth.

-Ishmael wants to try out as many roller coasters as possible, which includes extreme thrills, hence Ming and Ishmael cannot be together. He also wants to attend the magic show.

-Karine is very impatient and cannot wait in long lines which make splitting up her with Ming perfect as Ming cannot tolerate the extreme thrills and usually it's the extreme thrills that have long lines. She also wants to see the circus acrobatics show. And she has some allergy that makes her unable to eat hot dogs.

-Freya likes water rides and wants to spend time with friends. She also wants to have a souvenir to remember the trip.

1.2 Research motivation

-To develop a method to do scheduling while meeting the preferences of all friends

-To create a Mathematical Model to evaluate the scheduled cases enjoyment level

-To select the best trip schedule for the four group of friends to meet all preferences

Chapter 2 Proposed Methodology

2.1 Trip planning using Gantt chart

Gantt charts help in planning and scheduling projects by showing the duration and name of each process in an orderly and concise manner. The principle of how it works is very simple. Hence, Gantt charts is a powerful tool [1] to use in this project for scheduling the day at the park for the four friends. Excel is an excellent tool in creating Gantt Charts, hence, we made use of Excel to create all the Gantt charts in this project.

This section focuses on the components of a typical Gantt chart and the various processes involved making our own Gantt charts for the purposes of this investigation.

2.11 How we made our Gantt charts

The example below illustrates how we created Karine's personalized schedule. The Gantt charts of the individuals would explained in more detail later in this report.

Step 1: Making the timeline

Making the timeline is simple. We simply used Excel autofill to finish it.

9:00 AM	10:00 AM	11:00 AM	12:00 PM	1:00 PM	2:00 PM	3:00 PM	4:00 PM	5:00 PM	6:00 PM	7:00 PM	8:00 PM	9:00 PM

Step 2: Making the list of tasks

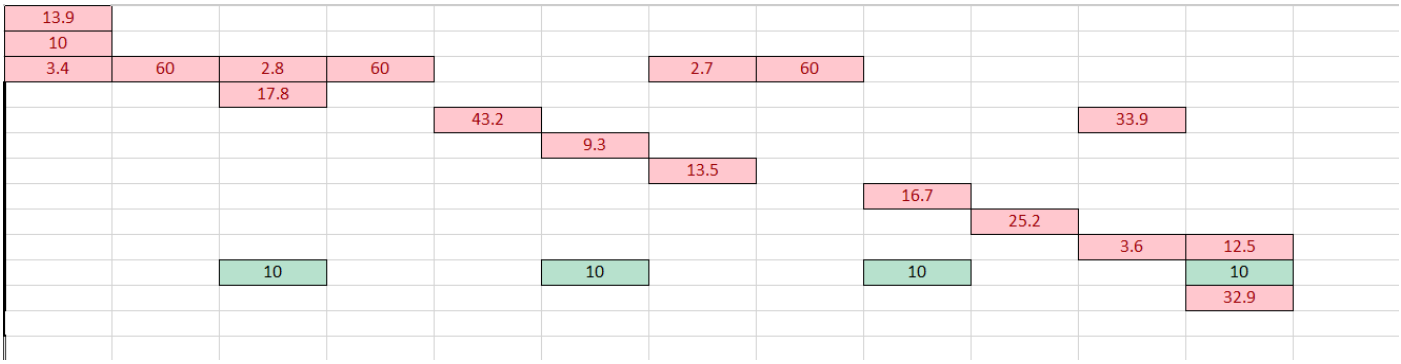
We inputted the task one by one into the vertical column and added a column beside the task column showing who should complete the task.

Dodecagon Rapids	Karine
Octagon Flume	Karine
Trapezoid Show	Karine
Ice Cream	Karine
Triangle Restaurant	Karine
Star Burst Carousel	Karine
Square Scramble	Karine
Rhombus Ride	Karine
Concave Train	Karine
Heptagon Coaster	Karine
WC	Karine
Gift Shop	Karine

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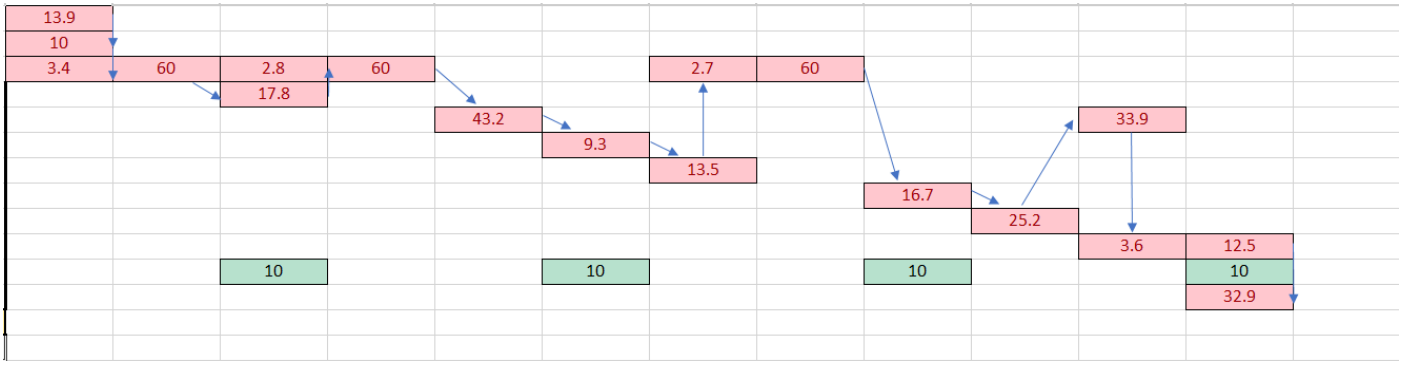
Step 3: Making the schedule

Next, we made the bars. Each bar also displays the duration of each activity. We colour coded the bars using Excel's conditional formatting function. The red bars are the main task while the green task is the toilet buffer times.



Step 4: Adding the arrows

Just having bars like was insufficient as we needed to show the friends the sequence of their activities. Hence, we simply added arrows that point towards the next activity to be completed.



2.12 Gantt chart analysis flowchart

Figure 1 is a flow chart of the process and steps to make the Gantt Chart. This flow chart demonstrates the process of making various different Gantt Charts for this project.

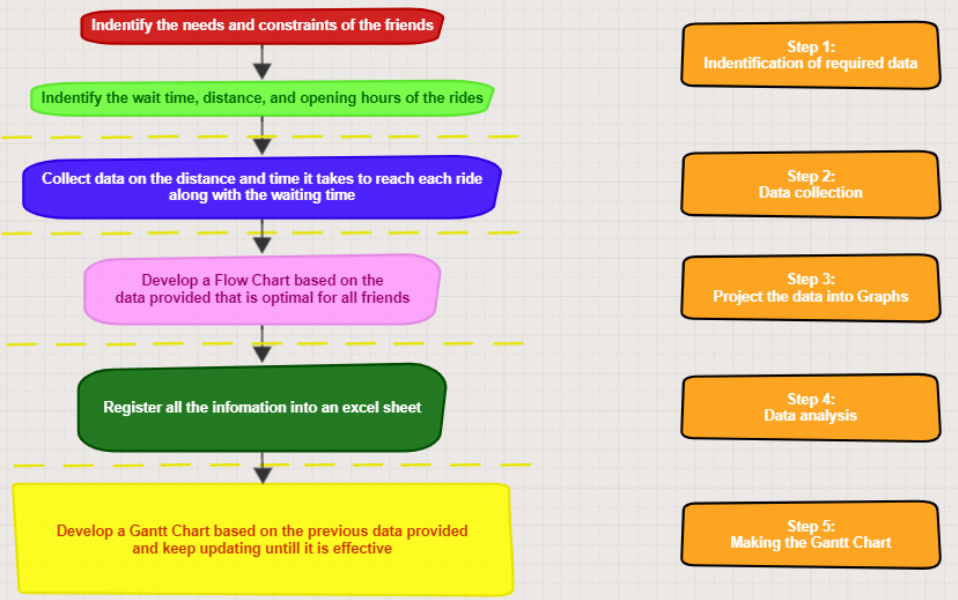


Figure 1 Scheduling by Gantt chart analysis flowchart

2.2 Variables

The variables we considered as shown in table 1:

Table 1 Mathematical model analysis variables definition

Variable	**Description**
H_i	Happiness score, i —-- team member
C_j	Coefficient of happiness, $C_j \in [-2,2]$
$t_{activity}$	Time spent on $activity$ (min)
t_{wait}	Waiting time (minutes)
t_{walk}	Walking time (minutes)
t_{buffer}	Buffer time (minutes)
α_k	$\alpha_k \in [0,0.6,1]$ $\alpha_k = 1$ if requirement is met $\alpha_k = 0$ if requirement is not met α_k = 0.6 is when the requirement stated by the question is not met. For example, separating the friends for more than 4 hrs a day.
D	Walking distance (cm)

2.3 Assumptions

Global assumptions

1. The friends' happiness can be evaluated by activity time, walking time, etc.
2. The group of friends enjoy activities such as having lunch and dinner and using the WC.
3. They cannot walk through each of the rides as we took the rides drawn in the map to be solid. For example, you cannot walk through the games stall.
4. The friends will enjoy the time devoted to activities more than the buffer time and they will prefer the waiting time over the walking time

Local assumptions

1. Approximate walking speed is 1.3m/s

Justification: Since the average walking speed of a human is 1.2~1.4m/s [2], we took the average of that number, 1.3m/s for the calculation of the time taken to move from each ride.

2. Time for toilet is about 10 to 20 mins

Justification: We have decided that the friends would go to the toilet during the time that they are eating, or also when there is extra buffer time. We assume that the friends take a maximum of 20 mins and a minimum of 5 mins to use the toilet. We did not consider the toilet waiting times.

3. Waiting time for the schedule is approximate and there is no change in an hour (such as 9a.m. to 10a.m.)

Justification: Although the chart given is not flexible, we have buffer time in case there is more waiting time than expected.

4. Play duration time is fixed for every activity

In this project we define buffer time differently. Buffer time refers to the room for error in the event of unexpected situations. If at the end of a certain activity, there is a surplus of allocated buffer time , the group of friends can decided how best to use it. For example, taking a group picture or going on a ride that they all enjoyed.

2.4 Steps for mathematical model development

1st stage → We are tasked to consider what the group of friends could experience in Polygon Paradise Park to get the most out of what they enjoy doing and avoid aspects they would find less pleasurable. We achieved this through research into both the park facilities and the individual preferences within the group. After going through this process, we plotted out the complete schedule for each of the 4 friends.

2nd stage → We are instructed to create a schedule which includes the group of friends spending the day together as a whole group. We are required to list down the activities, plan the timing for each zz

and find out which ones require the group to divide into 2 groups in order to maximise the enjoyment.

3rd stage → We are tasked to share our planned schedule with the friends which they would be using to guide them that day by writing them a letter.

4th stage → We need to think about how the schedule will differ from person to person, and whether or not the same process can be applied.

5th stage → We need to build a model to gauge whether the schedule is reasonable.

2.5 Proposed happiness mathematical model

$$\boldsymbol{H_i = \alpha_k(C_1 \times t_{wait} + C_2 \times t_{walk} + C_3 \times t_{activity} + C_4 \times t_{buffer})}$$

Deriving H

Our challenge is to calculate the happiness level of each person and add them together. We will gauge the happiness level by multiplying the Coefficient of happiness C_j and the time taken for waiting, walking, buffer and activities t_{wait} , t_{walk} , $t_{activity}$, t_{buffer} respectively, giving us the total level of happiness scored, denoted by H.

Deriving C_j

We will strive find the coefficient Cj describing each friend's happiness, with larger values of Cj indicating more happiness. This coefficient will be used to compare each friend's happiness level versus the activity he or she is carrying out. Our value of Cj is given by:

-2 points is for walking time $\boldsymbol{C_1}$

-1 points is for waiting time $\boldsymbol{C_2}$

+1 point is for the buffer time $\boldsymbol{C_3}$

+2 points is for the activity time $\boldsymbol{C_4}$

$\alpha = 0$ is when requirement is not met so the H will not be awarded for that group

$\alpha = 1$ is when requirement is met so that H will be awarded for that group

$\alpha = 0.6$ is when the friends stay together requirement stated by the MidMCM question is not met. For example, separating the friends for more than 4 hrs a day.

Where $C_j \in [-2,2]$ is a dimensionless value describing the friends' level of happiness throughout the day.

$$H = \sum_{i=1}^{4} \boldsymbol{H_i}$$

i=1⟶Ming,

i=2⟶Ishmael,

i=3⟶Karine,

i=4⟶Freya.

Chapter 3 Presentation of Model and Solution

3.1 Model data collection and simulation cases

Table 2 shows all the scheduling cases which include two members as one group, 4 members as one group, and only consider one member's requirement etc.

Table 2 Scheduling Cases

	9:00am~9:00pm
Case 1	Only meet Ming's (M) requirements
Case 2	Only meet Ishmael's (I) requirements
Case 3	Only meet Karine(K) requirements
Case 4	Only meet Freya(F) requirements

Case 5	Arrange M, I, K, F as one group		
Case 6	Split up into groups of two for up to 4 hours of the day		
	9:00am~1:00pm	1:00pm~5:00pm	5:00pm~9:00pm
Case 7	Group 1(M,I) Group 2 (K,F)	Group 1 (M, K) Group 2 (I, F)	Group 1 (M, F) Group 2 (K, I)
Case 8	Group 1 (M, K) Group 2 (I, F)	Group 1 (M, F) Group 2 (K, I)	Group 1 (M, I) Group 2 (K, F)
Case 9	Group 1 (M, F) Group 2 (K, I)	Group 1 (M, I) Group 2 (K, F)	Group 1 (M, K) Group 2 (I, F)

Table 3 – 5 shows the resources provided by MidMCM travel agency.

Table 3 Trapezoid Show Schedule

Time	**Show**	**Duration**
10 am	Circus Acrobatics	1 hour
12 pm	Circus Acrobatics	1 hour
2 pm	Magic Show	1 hour
4 pm	Circus Acrobatics	1 hour
6 pm	Magic Show	1 hour
8 pm	Firework Show	1 hour

Table 4 Ride information

Name of Ride/Activity	**Ride Type/Activity**	**Duration**
Heptagon Coaster	Extreme Thrill	2.5 min
Decagon Coaster	Extreme Thrill	3.0 min
Rhombus Ride	Extreme Thrill	4.0 min
Hexa-Swings	Medium Thrill	2.0 min
Pentagon Bumpers	Medium Thrill	5.0 min
Square Scramble	Medium Thrill	2.5 min
Concave Train	Basic	15.0 min
Starburst Carousel	Basic	3.0 min
Dodecagon Rapids	Water	10.0 min
Octagon Flume	Water	5.0 min

Table 5 Food Options

Name	**Types of Food**	**Hours of Operation**
Hot Dog Stand	Take Away	10:00 AM - 9:00 PM
Ice Cream Cart	Take Away	11:00 AM - 9:00 PM
Triangle Restaurant	Sit Down	9:00 AM - 8:00 PM

3.11 Polygon Paradise Park map data analysis

We first measured the distance between each of the rides on the map. The map is provided by MidMCM travel agency with a scale size of 1cm : 25m. Then we found out the average human walking speed to be 1.3m/s. Then using the data, we produced the formula to convert the distance in cm to the time it takes to walk the entire route in minutes.

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Table 5 Activity route distance in the map (cm)

	Concave Tr	Decagc	Dodecagon Rap	Heptagon Coast	Hexa-Swings	Octagon Flume	Pentagon Bump	Rhombus Ride	Square Scrambl	Star Burst Carou	Main entrance	WC	Ice Cream	Hot dogs	Triangle Restura	Gift Shop	Trapezoid Show	Games
Concave Train	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~
Decagon Coaste	2.7	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~
Dodecagon Rap	14.8	14.8	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~
Heptagon Coast	12.5	10.6	5.1	~	~	~	~	~	~	~	~	~	~	~	~	~	~	~
Hexa-Swings	2	3.6	16.1	11.6	~	~	~	~	~	~	~	~	~	~	~	~	~	~
Octagon Flume	12	8.5	15.7	11.2	12	~	~	~	~	~	~	~	~	~	~	~	~	~
Pentagon Bump	14	12.4	3	2.5	13.9	13.5	~	~	~	~	~	~	~	~	~	~	~	~
Rhombus Ride	16.2	14.8	2.7	6	16.4	15.8	4.5	~	~	~	~	~	~	~	~	~	~	~
Square Scrambl	7.5	5	13.5	9.2	8.3	3.7	11.3	13.6	~	~	~	~	~	~	~	~	~	~
Star Burst Carou	9.7	7.1	11.6	7	9.1	4.2	9.3	11.6	3.1	~	~	~	~	~	~	~	~	~
Main Entrance	13.1	11.4	12.3	7.5	13.2	7.5	9.8	12.3	7.7	5	~	~	~	~	~	~	~	~
WC	7.1	5.3	11.7	7	6.6	6	9.5	11.7	2.7	3.2	7.8	~	~	~	~	~	~	~
Ice Cream	9	6.1	13.9	9.4	9.6	2.5	11.7	14	1.2	3	7.2	3.6	~	~	~	~	~	~
Hot Dogs	13.1	11.3	3.8	2.8	12.7	12.2	2.7	4	10	8	8.7	8.1	10.4	~	~	~	~	~
Triangle Restaur	12.3	9.6	15.7	11.2	13.1	1.8	13.3	15.6	4.8	4.1	6.9	6.6	3.5	12.1	~	~	~	~
Gift Shop	13.3	10	13.3	8.9	12.1	4.9	10.9	13.3	5.7	3.1	2.7	6.3	4.9	9.8	4.3	~	~	~
Trapezoid Show	11.5	9.8	8.6	3	11.2	10.5	5.4	8.5	8.5	6.4	4.7	6.7	8.8	5.1	9.9	6.4	~	~
Games	7.2	5.7	9.2	4.7	7	7.6	7.1	9.4	5	3.6	6	2.7	5.6	5.8	7.6	6.8	4.3	~

Table 5 shows the shortest distance measured in centimetres between each of the locations in the park. The data was measured directly from the map as it is drawn to scale. When measuring, we considered the fact that some of the routes have obstacles in between them which need circumventing.

Table 6 Walking time between different rides (minutes)

	Concave Tr	Decagc	Dodecagon Rap	Heptagon Coast	Hexa-Swings	Octagon Flume	Pentagon Bump	Rhombus Ride	Square Scrambl	Star Burst Carou	Main entrance	WC	Ice Cream	Hot dogs	Triangle Restura	Gift Shop	Trapezoid Show	Games
Concave Train		0.9	4.7	4.0	0.6	3.8	4.5	5.2	2.4	3.1	4.2	2.3	2.9	4.2	3.9	4.3	3.7	2.3
Decagon Coaste	0.9		4.7	3.4	1.2	2.7	4.0	4.7	1.6	2.3	3.7	1.7	2.0	3.6	3.1	3.2	3.1	1.8
Dodecagon Rap	4.7	4.7		1.6	5.2	5.0	1.0	0.9	4.3	3.7	3.9	3.8	4.5	1.2	5.0	4.3	2.8	2.9
Heptagon Coast	4.0	3.4	1.6		3.7	3.6	0.8	1.9	2.9	2.2	2.4	2.2	3.0	0.9	3.6	2.9	1.0	1.5
Hexa-Swings	0.6	1.2	5.2	3.7		3.8	4.5	5.3	2.7	2.9	4.2	2.1	3.1	4.1	4.2	3.9	3.6	2.2
Octagon Flume	3.8	2.7	5.0	3.6	3.8		4.3	5.1	1.2	1.3	2.4	1.9	0.8	3.9	0.6	1.6	3.4	2.4
Pentagon Bump	4.5	4.0	1.0	0.8	4.5	4.3		1.4	3.6	3.0	3.1	3.0	3.8	0.9	4.3	3.5	1.7	2.3
Rhombus Ride	5.2	4.7	0.9	1.9	5.3	5.1	1.4		4.4	3.7	3.9	3.8	4.5	1.3	5.0	4.3	2.7	3.0
Square Scrambl	2.4	1.6	4.3	2.9	2.7	1.2	3.6	4.4		1.0	2.5	0.9	0.4	3.2	1.5	1.8	2.7	1.6
Star Burst Carou	3.1	2.3	3.7	2.2	2.9	1.3	3.0	3.7	1.0		1.6	1.0	1.0	2.6	1.3	1.0	2.1	1.2
Main Entrance	4.2	3.7	3.9	2.4	4.2	2.4	3.1	3.9	2.5	1.6		2.5	2.3	2.8	2.2	0.9	1.5	1.9
WC	2.3	1.7	3.8	2.2	2.1	1.9	3.0	3.8	0.9	1.0	2.5		1.2	2.6	2.1	2.0	2.1	0.9
Ice Cream	2.9	2.0	4.5	3.0	3.1	0.8	3.8	4.5	0.4	1.0	2.3	1.2		3.3	1.1	1.6	2.8	1.8
Hot Dogs	4.2	3.6	1.2	0.9	4.1	3.9	0.9	1.3	3.2	2.6	2.8	2.6	3.3		3.9	3.1	1.6	1.9
Triangle Restaur	3.9	3.1	5.0	3.6	4.2	0.6	4.3	5.0	1.5	1.3	2.2	2.1	1.1	3.9		1.4	3.2	2.4
Gift Shop	4.3	3.2	4.3	2.9	3.9	1.6	3.5	4.3	1.8	1.0	0.9	2.0	1.6	3.1	1.4		2.1	2.2
Trapezoid Show	3.7	3.1	2.8	1.0	3.6	3.4	1.7	2.7	2.7	2.1	1.5	2.1	2.8	1.6	3.2	2.1		1.4
Games	2.3	1.8	2.9	1.5	2.2	2.4	2.3	3.0	1.6	1.2	1.9	0.9	1.8	1.9	2.4	2.2	1.4	

In table 6, we have been referring to input our walking time into our charts. We have assumed that the average walking speed of the friends to be 1.3m/s as it is the average of the normal walking speed of humans, between 1.2m/s and 1.4m/s. Hence, we inputted the values in the distance table into this formula:

$$t_w = \frac{((D*25)/1.3}{60}$$

-Where tw is the values in the 'wait' timetable and D is the values in the 'Distance' table. In the formula, D is multiplied with twenty-five as the map is drawn to a scale of 1 cm representing 25m. Next, we divided it by the average human walking speed (which is 1.3 m/s) and convert it to the minutes needed to walk between locations. Hence, we divided by 60 to find the minutes needed to walk between the locations.

3.12 Waiting time analysis

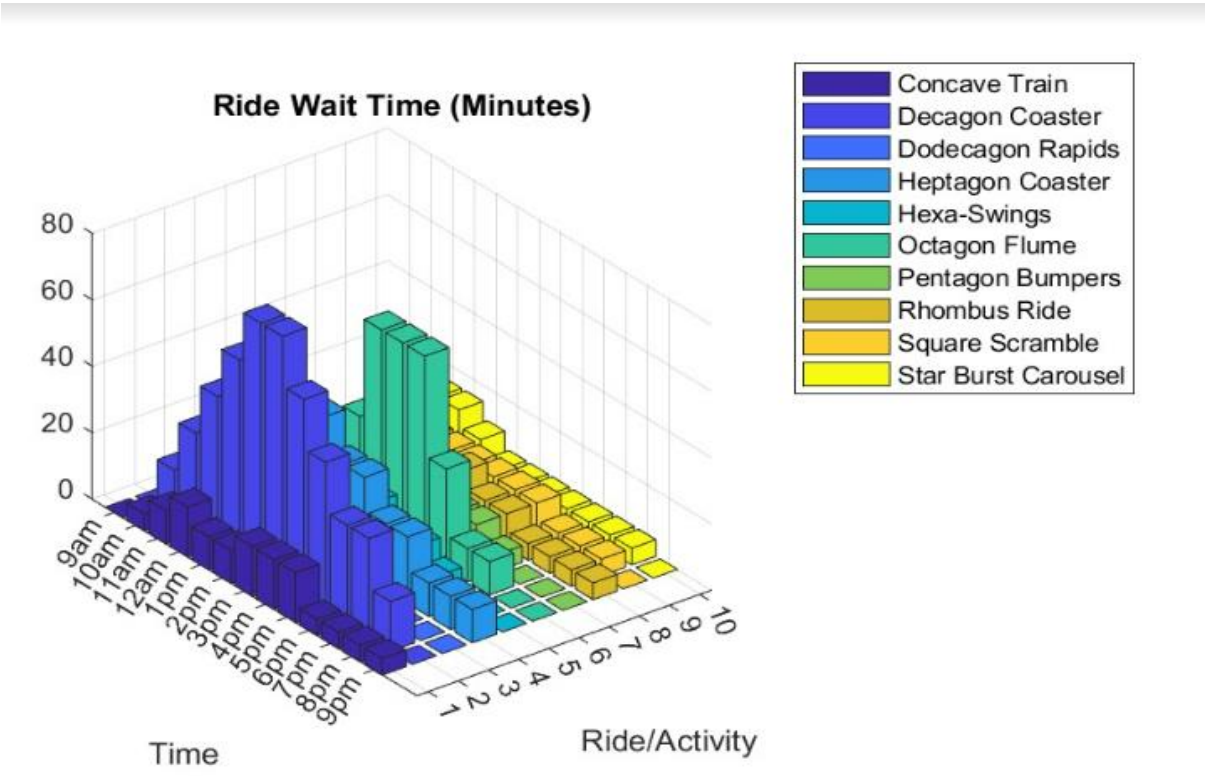


Figure 2 Ride Waiting Time

Figure 2 is a 3D Graph made based on the data of the ten different waiting times for each ride. This Graph was made to make it easier for the friends refer to and accurately pinpoint the waiting times immediately.

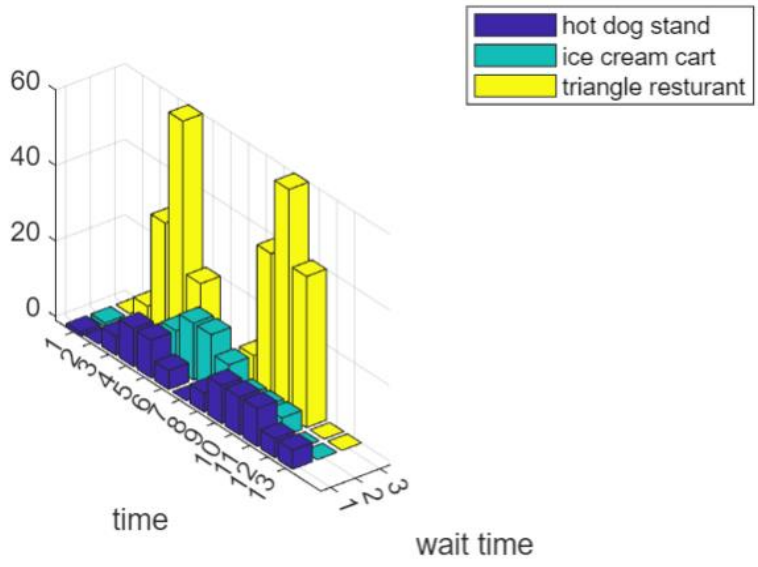
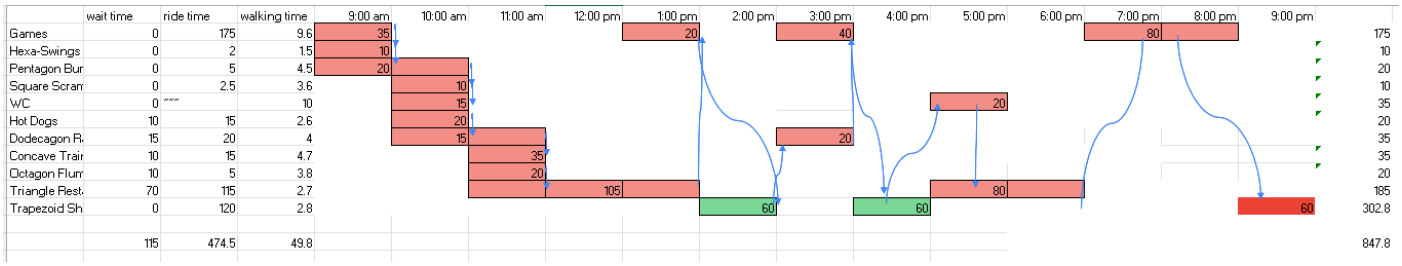


Figure 3 Food waiting time

Figure 3 is a 3D Graph made based on the data of the food waiting times. Just like the previous Graph, it was made from Matlab and used for the purpose of easily ascertaining the waiting times for the food.

3.2 Scheduling analysis by Gantt chart

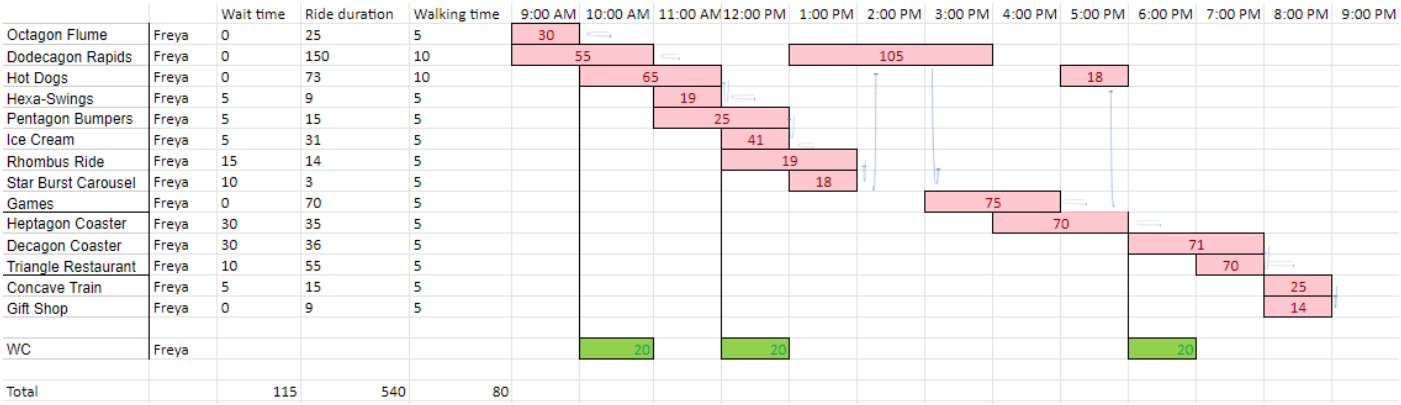
3.21 Scheduling only consider single team member requirement



	wait time	ride time	walking time	9:00 am	10:00 am	11:00 am	12:00 pm	1:00 pm	2:00 pm	3:00 pm	4:00 pm	5:00 pm	6:00 pm	7:00 pm	8:00 pm	9:00 pm	
Games	0	175	9.6	35				20		40				80			175
Hexa-Swings	0	2	1.5	10													10
Pentagon Bur	0	5	4.5	20													20
Square Scran	0	2.5	3.6		10												10
WC	0	"""	10		15							20					35
Hot Dogs	10	15	2.6		20												20
Dodecagon R	15	20	4		15					20							35
Concave Trair	10	15	4.7			35											35
Octagon Flum	10	5	3.8			20											20
Triangle Rest	70	115	2.7				105					80					185
Trapezoid Sh	0	120	2.8					60			60				60		302.8
	115	474.5	49.8														847.8

Figure 4 Scheduling by Gantt chart on Ming's preference.

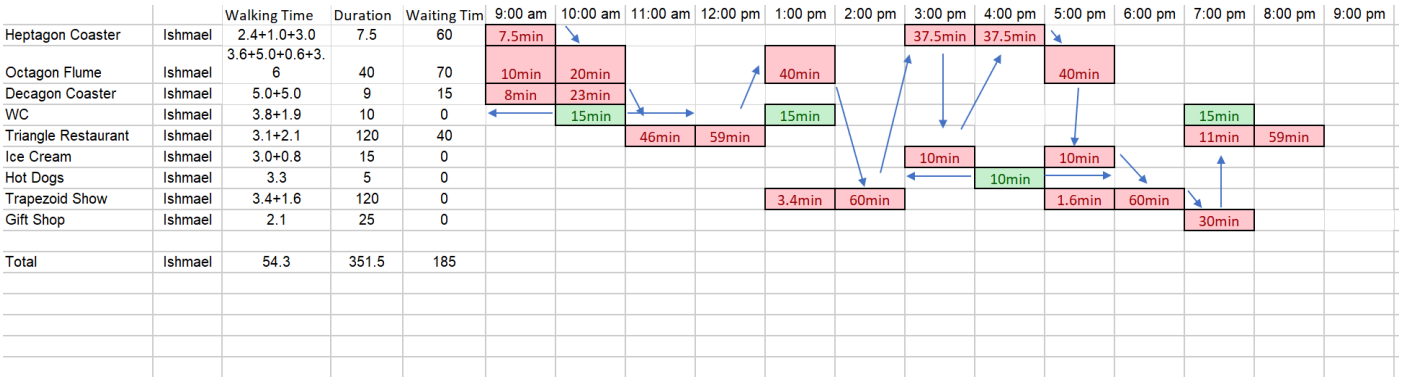
As Ming likes to play at game stalls, in Figure 4 he will have ample time to win some prizes. This chart includes no extreme thrills. However, there are limitations for this chart as Ming doesn't really have that many restrictions, thus there are some rides that he did not state in his preferences.



		Wait time	Ride duration	Walking time	9:00 AM	10:00 AM	11:00 AM	12:00 PM	1:00 PM	2:00 PM	3:00 PM	4:00 PM	5:00 PM	6:00 PM	7:00 PM	8:00 PM	9:00 PM
Octagon Flume	Freya	0	25	5	30												
Dodecagon Rapids	Freya	0	150	10		55			105								
Hot Dogs	Freya	0	73	10		65							18				
Hexa-Swings	Freya	5	9	5			19										
Pentagon Bumpers	Freya	5	15	5			25										
Ice Cream	Freya	5	31	5				41									
Rhombus Ride	Freya	15	14	5				19									
Star Burst Carousel	Freya	10	3	5					18								
Games	Freya	0	70	5							75						
Heptagon Coaster	Freya	30	35	5								70					
Decagon Coaster	Freya	30	36	5										71			
Triangle Restaurant	Freya	10	55	5											70		
Concave Train	Freya	5	15	5												25	
Gift Shop	Freya	0	9	5												14	
WC	Freya					20		20						20			
Total		115	540	80													

Figure 5 Scheduling by Gantt chart on Freya's preference.

Figure 5 is the representation of the Gantt Chart created for Freya. As Freya likes to spend time on water rides we have given her a chance to cool off on the morning rides where there is no waiting time on the water rides. Freya also likes to go with the flow so she can ride on popular rides without much waiting time.

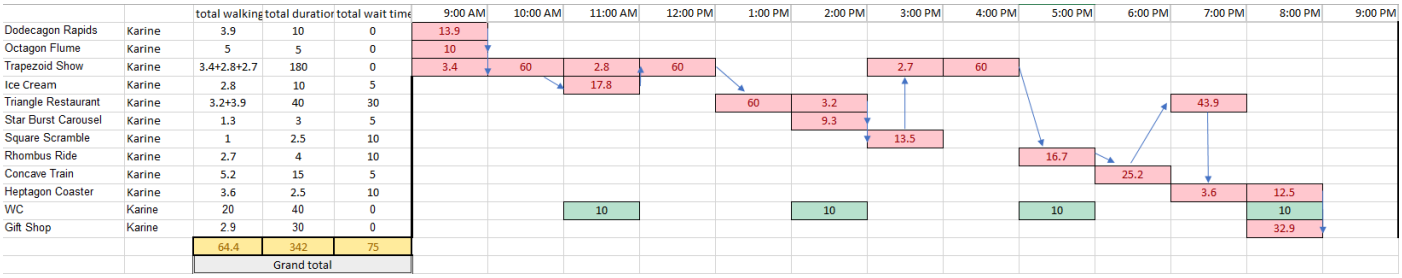


		Walking Time	Duration	Waiting Tim	9:00 am	10:00 am	11:00 am	12:00 pm	1:00 pm	2:00 pm	3:00 pm	4:00 pm	5:00 pm	6:00 pm	7:00 pm	8:00 pm	9:00 pm
Heptagon Coaster	Ishmael	2.4+1.0+3.0	7.5	60	7.5min						37.5min	37.5min					
Octagon Flume	Ishmael	3.6+5.0+0.6+3.6	40	70	10min	20min			40min				40min				
Decagon Coaster	Ishmael	5.0+5.0	9	15	8min	23min											
WC	Ishmael	3.8+1.9	10	0		15min			15min						15min		
Triangle Restaurant	Ishmael	3.1+2.1	120	40			46min	59min							11min	59min	
Ice Cream	Ishmael	3.0+0.8	15	0							10min		10min				
Hot Dogs	Ishmael	3.3	5	0								10min					
Trapezoid Show	Ishmael	3.4+1.6	120	0					3.4min	60min			1.6min	60min			
Gift Shop	Ishmael	2.1	25	0											30min		
Total	Ishmael	54.3	351.5	185													

Figure 6 Scheduling by Gantt chart on Ishmael's preference.

Figure 6 is the Gantt chart representation created for Ishmael. He loves roller coasters and wants to ride as many as possible during the day. Thus, he gets to spend majority of the time on the Heptagon Coaster and

Decagon coaster, which are the 2 roller coaster rides. At 2pm and 6pm, he will attend the magic show in Trapezoid Show since he enjoys performing card tricks.

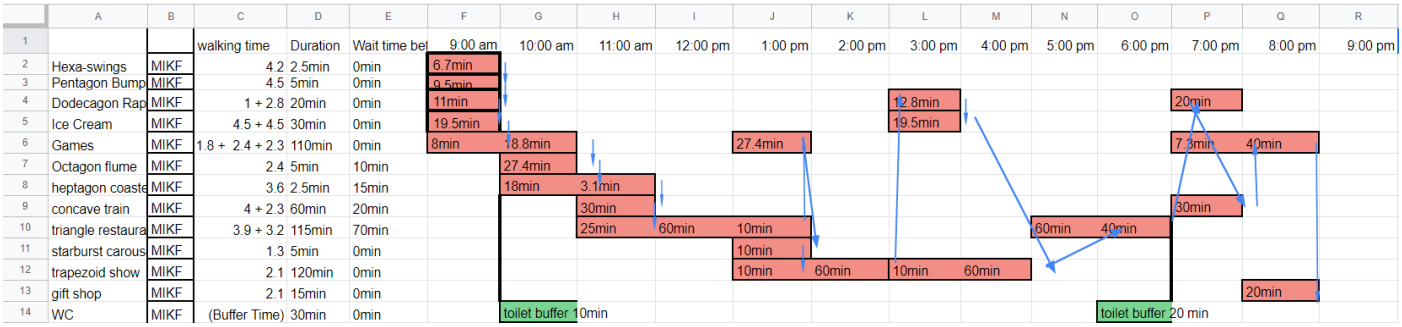


		total walking	total duration	total wait time	9:00 AM	10:00 AM	11:00 AM	12:00 PM	1:00 PM	2:00 PM	3:00 PM	4:00 PM	5:00 PM	6:00 PM	7:00 PM	8:00 PM	9:00 PM
Dodecagon Rapids	Karine	3.9	10	0	13.9												
Octagon Flume	Karine	5	5	0	10												
Trapezoid Show	Karine	3.4+2.8+2.7	180	0	3.4	60	2.8	60			2.7	60					
Ice Cream	Karine	2.8	10	5			17.8										
Triangle Restaurant	Karine	3.2+3.9	40	30					60	3.2							
Star Burst Carousel	Karine	1.3	3	5						9.3							
Square Scramble	Karine	1	2.5	10							13.5						
Rhombus Ride	Karine	2.7	4	10									16.7				
Concave Train	Karine	5.2	15	5										25.2			
Heptagon Coaster	Karine	3.6	2.5	10											43.9		
WC	Karine	20	40	0			10			10			10		3.6	12.5	
Gift Shop	Karine	2.9	30	0												10	
		64.4	342	75												32.9	
		Grand total															

Figure 7 Scheduling by Gantt chart on Karine's preference.

Figure 7 is the Gantt chart for Karine. Since Karine likes Circus Acrobatics, she would go for all the Circus Acrobatics shows at 10:00 am, 12:00 pm and 4:00 pm. Karine would also have her lunch and dinner at 1:00 pm and 7:00 pm. Karine would go for the water rides in the morning to cool off as the water rides would have long waiting times at noon, especially Octagon flume. After the Circus Acrobatics which is after the water rides, Karine would get ice cream as she cannot eat hotdogs. From 1:00 pm to 6:00 pm, Karine would go for the rides with the shortest wait times as she is impatient. At the end of the day, she would go for the Heptagon Coaster before visiting the gift shop.

3.22 Scheduling for four members in one group



		walking time	Duration	Wait time bet	9:00 am	10:00 am	11:00 am	12:00 pm	1:00 pm	2:00 pm	3:00 pm	4:00 pm	5:00 pm	6:00 pm	7:00 pm	8:00 pm	9:00 pm
Hexa-swings	MIKF	4.2	2.5min	0min	6.7min												
Pentagon Bump	MIKF	4.5	5min	0min	9.5min												
Dodecagon Rap	MIKF	1 + 2.8	20min	0min	11min						12.8min				20min		
Ice Cream	MIKF	4.5 + 4.5	30min	0min	19.5min						19.5min						
Games	MIKF	1.8 + 2.4 + 2.3	110min	0min	8min	18.8min			27.4min						7.3min	40min	
Octagon flume	MIKF	2.4	5min	10min		27.4min											
heptagon coaste	MIKF	3.6	2.5min	15min		18min	3.1min										
concave train	MIKF	4 + 2.3	60min	20min			30min								30min		
triangle restaura	MIKF	3.9 + 3.2	115min	70min			25min	60min	10min				60min	40min			
starburst carous	MIKF	1.3	5min	0min					10min								
trapezoid show	MIKF	2.1	120min	0min					10min	60min	10min	60min					
gift shop	MIKF	2.1	15min	0min												20min	
WC	MIKF	(Buffer Time)	30min	0min		toilet buffer	10min							toilet buffer	20 min		

Figure 8 Scheduling by Gantt chart on MIKF's preference.

Figure 8 is the flow chart designed for the group when they were all together with all the limitations. The activities planned are to minimise the number of extremes thrills for M and try to go on every other ride. In the figure there are also activities designed for F in terms of water activities. This way Ishmael will get to try out extreme rollercoasters but without Ming getting too sick. Also, Karine will not eat hot dogs so instead they will eat ice cream. This is the best plan for them satisfying most of the requirements.

3.23 Scheduling for pairs and four members per group

Before we show random groups' Gantt Chart, we will first explain our planning process. Instead of blindly diving into making the Gantt Chart, we first made an effective Flow Chart as shown in figure 9. Figure 9 demonstrates the planning of how the friends would carry out their activities during their day at the park, ensuring maximum possible happiness levels. The organisation of this would be finalised later in the Gantt chart. The flow chart was also supposed to be a way for us to rapidly generate ideas as a starting point. From the picture below, it is easy to see the route planned out for them and an average of how long it will take for them to get there. This was a crucial part of our planning as it is the foundation to the Gantt Chart.

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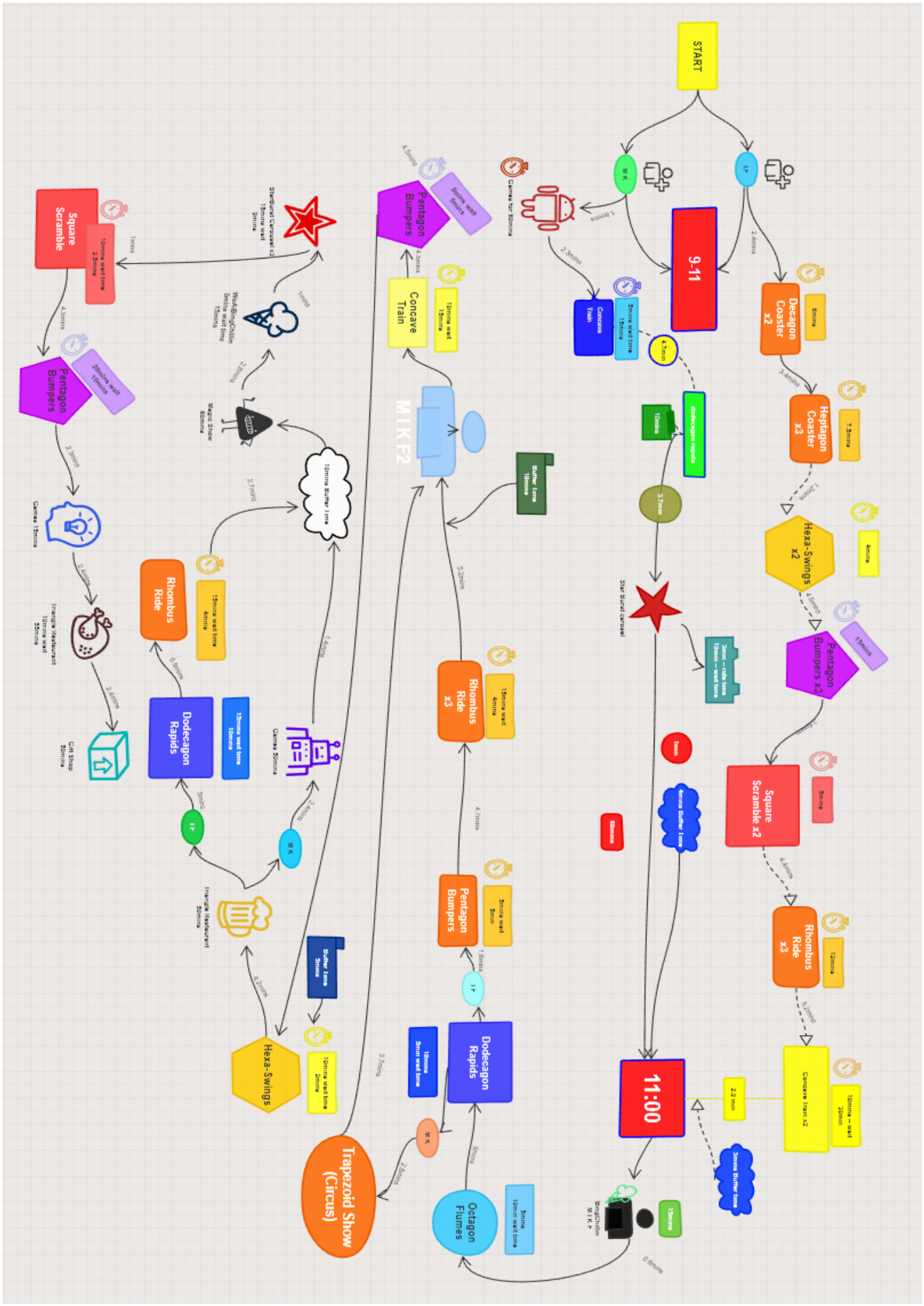
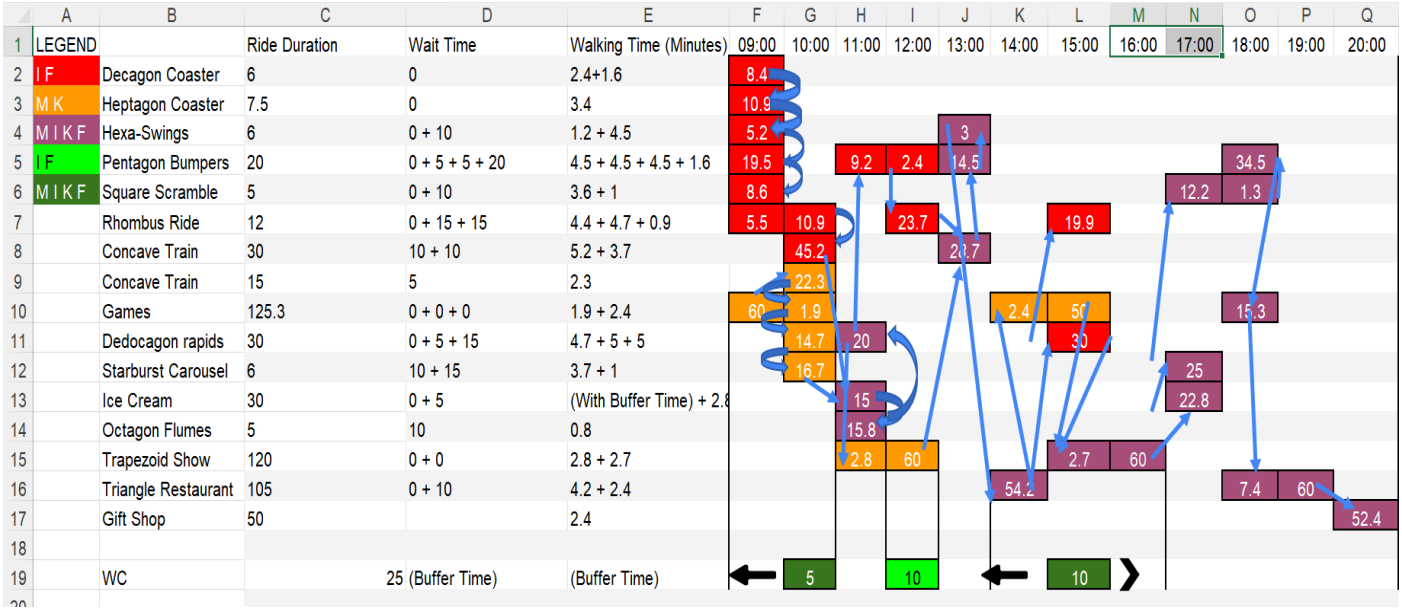


Figure 9 Trip flowchart (Case 6)



	LEGEND		Ride Duration	Wait Time	Walking Time (Minutes)
2	I F	Decagon Coaster	6	0	2.4+1.6
3	M K	Heptagon Coaster	7.5	0	3.4
4	M I K F	Hexa-Swings	6	0 + 10	1.2 + 4.5
5	I F	Pentagon Bumpers	20	0 + 5 + 5 + 20	4.5 + 4.5 + 4.5 + 1.6
6	M I K F	Square Scramble	5	0 + 10	3.6 + 1
7		Rhombus Ride	12	0 + 15 + 15	4.4 + 4.7 + 0.9
8		Concave Train	30	10 + 10	5.2 + 3.7
9		Concave Train	15	5	2.3
10		Games	125.3	0 + 0 + 0	1.9 + 2.4
11		Dedocagon rapids	30	0 + 5 + 15	4.7 + 5 + 5
12		Starburst Carousel	6	10 + 15	3.7 + 1
13		Ice Cream	30	0 + 5	(With Buffer Time) + 2.8
14		Octagon Flumes	5	10	0.8
15		Trapezoid Show	120	0 + 0	2.8 + 2.7
16		Triangle Restaurant	105	0 + 10	4.2 + 2.4
17		Gift Shop	50		2.4
18					
19		WC		25 (Buffer Time)	(Buffer Time)

Figure 10 Gantt chart (Case 6)

Figure 10 is case 6 of the Gantt Chart. There is a legend on the left side to show which colour signifies who (M = Ming. I = Ishmael. K = Karine. F = Freya) The numbers in the rectangular cells show the amount of time in minutes taken on the rides (including walking and wait time) and there is a specific column for the time it takes to walk to each destination.

Key Features

We added different ‘floating’ bars to represent flexible timings that can be adjusted, such as going to the toilet and also buffer timings just in case the friends decided to take a picture, buy a snack, and so on. We also added a small legend bar to ensure minimal confusion and added the approximate timings of walking to each ride. As our Gannt chart was hard to read initially, we decided to add alternating colours to each column to make it more accessible.

Challenges And Solutions

As our flow chart specified that the group of friends had to split up for a while then proceed back together, we had to figure out how to reflect this into our Gantt Chart. We assumed the timings of walking from one ride to another to be five minutes no matter what. We realised however that this was too inaccurate.

As is demonstrated by the final Gantt Chart, we colour coded each of the separate groups to simplify data input of the friends splitting up and regrouping again.

Instead of using inaccurate assumptions, we instead hand calculated the distance from every ride, stall, and restaurant and then converted it into an average time in minutes. This enabled our chart to be as accurate as possible for the friends to use.

3.24 Scheduling for two members per group

		total walking	total duration	total wait time	9:00 AM	10:00 AM	11:00 AM	12:00 PM	1:00 PM
Concave Train	Ming & Ishmael	4.2+2.3	30	5	19.2	7.3	15		
Star Burst Carousel	Ming & Ishmael	3.1	3	0	6.1				
Pentagon Bumpers	Ming & Ishmael	3+4.3	10	5	8			14.3	
Games	Ming & Ishmael	2.3	67	0	26.7	42.6			
Triangle Restaurant	Ming & Ishmael	3.9	30	30			43.9	20	
Dodecagon Rapids	Ming & Ishmael	1	10	10				21	
WC	Ming & Ishmael	~	~	~		30			
		17.6	150	50					
Octagon Flume	Karine & Freya	2.4	5	0	7.4				
Dodecagon Rapids	Karine & Freya	5	10	0	15				
Heptagon Coaster	Karine & Freya	1.6	2.5	0	4.1				
Decagon Coaster	Karine & Freya	3.4	3	0	6.4				
Rhombus Ride	Karine & Freya	0.9	4	0	4.9				
Trapezoid Show	Karine & Freya	2.7+1.4	120	0	2.7	60	1.4	60	
Games	Karine & Freya	1.4	57.2	0			58.6		
WC	Karine & Freya	~	~	~			20		
		18.8	201.7	0					

		total walking	total duration	total wait time	1:00 PM	2:00 PM	3:00 PM	4:00 PM	5:00 PM
Pentagon Bumpers	Ming & Karine	4.5	10	10	24.5				
Concave Train	Ming & Karine	4.5	15	10	29.5				
Triangle Restaurant	Ming & Karine	3.9	60	0	3.9	60			
Star Burst Carousel	Ming & Karine	1.3	3	5			9.3		
Games	Ming & Karine	1.2	32	0			33.2		
Trapezoid Show	Ming & Karine	1.4	60	0			1.4	60	
WC	Ming & Karine	~	~	~		20			
		16.8	180	25					
Dodecagon Rapids	Ishmael & Freya	5.2	35	15	55.2				
Trapezoid Show	Ishmael & Freya	2.8	60	0	2.8	60			
Triangle Restaurant	Ishmael & Freya	3.2	56.8	0			60		
Rhombus Ride	Ishmael & Freya	5	4	10				19	
Concave Train	Ishmael & Freya	5.2	15	15				35.2	
WC	Ishmael & Freya	~	~	~			20		
		21.4	170.8	40					

		total walking	total duration	total wait time	5:00 PM	6:00 PM	7:00 PM	8:00 PM	9:00 PM
Dodecagon Rapids	Ming & Freya	5.2	30	15	50.2				
Octagon Flume	Ming & Freya	5	10	20		35			
Games	Ming & Freya	2.4	40	0		12.4	30		
Triangle Restaurant	Ming & Freya	2.4	60	10			30	42.4	
WC	Ming & Freya	~	~	~			20		
		15	140	45					
Pentagon Bumpers	Karine & Ishmael	4.5	5	10	19.5				
Dodecagon Rapids	Karine & Ishmael	1	10	5	16				
Rhombus Ride	Karine & Ishmael	0.9	4	10	14.9				
Ice Cream	Karine & Ishmael	4.5	30	5	9.6	29.9			
Octagon Flume	Karine & Ishmael	0.8	5	10		10.8	5		
Hexa-Swings	Karine & Ishmael	3.8	2	0			5.8		
Games	Karine & Ishmael	2.2	38	0			40.2		
Triangle Restaurant	Karine & Ishmael	2.4	60	0			2.4	60	
WC	Karine & Ishmael	~	~	~	19.3			20	
		20.1	154	40					

Figure 11 Schedule for a group of 2(Case 7)

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Figure 11 is the representation of the Gantt Chart for a separate group of 2 people in a time span of 4 hours each. Since for this case the friends can split up in pairs, we have made sure that they would remain as happy as possible. Ming likes playing games and will suffer from motion sickness if he goes on extreme or medium thrill rides after eating. For each of the possible cases involving Ming, we have therefore allocated a minimum time of 40mins for games, and we ensure that he avoids extreme and medium thrill rides after eating. For Ishmael, who likes riding roller coasters and wants to watch the magic shows, we made sure that he would at least be able to watch the magic show and enjoy all the rides in the park. For Karine, who does not like waiting, does not eat hotdogs, and wants to watch the acrobatics circus shows, we ensured that she had the least waiting time possible and included one entire 4-hour slot devoid of any waiting time at all. We also gave her time to watch the circus show that she wanted and made sure she did not eat hotdogs. For Freya, who likes going on water rides to cool off, likes to go with the flow, and loves spending time with her friends, we made sure that she was able to enjoy her time on the water rides, and go on other popular rides.

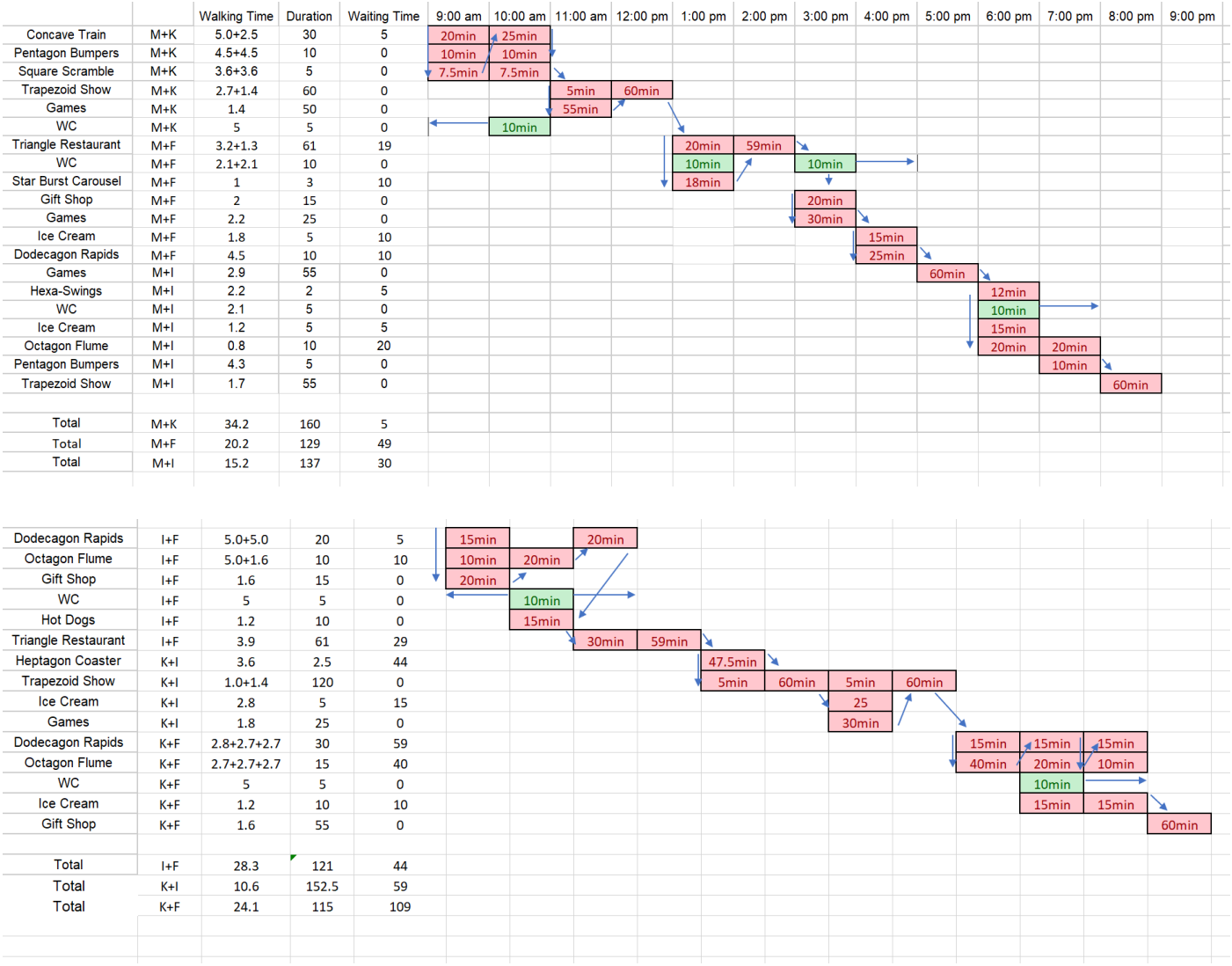


Figure 12 Schedule for a group of 2 (Case 8)

Figure 12 is the Gantt chart produced for groups. For each of the friends maximise levels of enjoyment, they are split up into two groups, and they will take turns to pair up every 4 hours. The activities are planned such

that everyone can meet their requirements. M loves to play games, but he gets motion sick on fast moving rides, especially after eating. This means that he cannot go on Extreme Thrill rides and no Medium Thrill for him after eating. Ishmael loves roller coasters and hence he is scheduled to take as many roller coaster rides as possible during the day. We also made sure that he has time to attend the Trapezoid Show as he enjoys seeing the magic show. Karine hates waiting in long lines so activities for her are optimised such that she will have as little wait time as possible. He also wants to attend the circus acrobatics show and does not eat hot dogs. Freya likes water rides so she will spend most of the time on water activities. We also took note and provide her with time to visit the Gift Shop.

		total walking	total duration	total wait time	9:00 AM	10:00 AM	11:00 AM	12:00 PM	1:00 PM
Dodecagon Rapids	Ming & Freya	3.9	30	0	33.9				
Octagon Flume	Ming & Freya	5	22	0	23	4			
Games	Ming & Freya	2.4+1.8	65	0		26.4		42.8	
Triangle Restauran	Ming & Freya	2.4	40	5		7.4	40		
Ice cream	Ming & Freya	1.1	10	10			20	1.1	
WC	Ming & Freya	5	10	0		10		10	
		21.6	177	15					
Heptagon Coaster	Karine & Ishmael	2.4	7.5	0	9.9				
Decagon Coaster	Karine & Ishmael	3.4	9	0	12.4				
Octagon Flume	Karine & Ishmael	2.7	15	0	17.7				
Trapezoid Show	Karine & Ishmael	3.4+1.7	120	0	3.4	60	1.7	60	
Ice cream	Karine & Ishmael	2.8	10	10			22.8		
Pentagon Bumpers	Karine & Ishmael	3.8	5	5			13.8		
WC	Karine & Ishmael	10	20	0	10		10		
		30.2	186.5	15					

		total walking	total duration	total wait time	1:00 pm	2:00 pm	3:00 pm	4:00 pm	5:00 pm
Concave Train	Ming & Ishmael	5.2	30	20	55.2				
Trapezoid Show	Ming & Ishmael	3.7	60	0	3.7	60			
Games	Ming & Ishmael	1.4	48.5	0			49.9		
Pentagon Bumpers	Ming & Ishmael	2.3	15	30				47.3	
WC	Ming & Ishmael	10	20	0			10	10	
		22.6	173.5	50					
Dodecagon Rapids	Karine & Freya	5.2+3.7	30	45	55.2		28.7		
Games	Karine & Freya	2.9	52	0	4.8	50.1			
Star Burst Carousel	Karine & Freya	1.2	6	10			17.2		
Trapezoid Show	Karine & Freya	2.8	60	0			2.8	60	
WC	Karine & Freya	10	20	0		10	10		
		25.8	168	55					

		total walking	total duration	total wait time	5:00 pm	6:00 pm	7:00 pm	8:00 pm	9:00 pm
Dodecagon Rapids	Ming & Karine	5.2	20	10	35.2				
Games	Ming & Karine	2.9	48.6	0	2.9	48.6			
Triangle Restaurant	Ming & Karine	2.4	50	10		2.4	60		
Gift Shop	Ming & Karine	1.4	58.6	0				60	
WC	Ming & Karine	10	20	0	10		10		
		21.9	197.2	20					
Dodecagon Rapids	Ishmael & Freya	5.2	10	5	20.2				
Heptagon Coaster	Ishmael & Freya	1.6	2.5	20	24.1				
Trapezoid Show	Ishmael & Freya	1	60	0	1	60			
Triangle Restaurant	Ishmael & Freya	3.2	56.8	0			60		
Gift Shop	Ishmael & Freya	1.4	48.6	0				50	
WC	Ishmael & Freya	5	10	0	10			10	
		17.4	187.9	25					

Figure 13 Schedule for a group of 2 (Case 9)

Figure 13 is the Gantt chart for separate groups of 2 people in the timespan of 4 hours. Ming has motion sickness and likes to play games; hence, Ming does not take any extreme thrills and has at least 30 minutes of game time no matter what group he is in. Freya likes water rides, thus, she managed to have her water rides in all three of her Gantt charts. She also gets to buy a souvenir at the end of the day. Karine is impatient and wants to watch all the Circus Acrobatics, so, we made sure he goes for all the Circus Acrobatics shows at that time span. We also made sure Karine's total waiting time does not exceed 35 minutes. We also made Karine eat ice cream as he could not eat hotdogs. Ishmael wants to watch all the magic shows and ride many roller coasters. So, we made sure he goes for all the magic shows and he gets to ride his roller coasters except for the Gantt chart where he is with Ming as Ming has motion sickness.

3.3 Happiness mathematical model assess all scheduling cases

As shown in table 6, its summary of all scheduling cases by the happiness mathematical model, according to the above analysis case 6 achieves the highest score. This is the optimal case, meeting everyone's requirements and making everyone happy. This case also satisfied the requirement stated in the question by separating group members for a total of 4 hrs time.

From Case 1 to Case 4 is where only one person's preferences are met, and it does not satisfy the other three people's needs, thus its H score is lower.

Case 5 is where there is completely no separation at all demonstrates how hard it is to make everyone happy due to all the limitations given. Hence it did not get a high happiness score. From the mathematical model assessment score can be seen, H score for case 5 is not higher than case 6.

Case 7- 9 are the cases that did not meet the requirements as they exceeded 4hrs of separation time. Thus, their H score is lower.

Thus, from the above analysis, we selected case 6 as the best schedule enabling maximum level of enjoyment for the group of friends.

Table 6 Happiness mathematical model analysis for all cases

	Waiting Time (Minutes)	Activity Time (Minutes)	Walking Time (Minutes)	Buffer Time (Minutes)	Happiness Scores
Case 1	115	474.5	49.5	60	434.4
Case 2	185	351.5	54.3	129.2	323.2
Case 3	45	342	64.4	268.6	467.3
Case 4	115	540	49.8	15.2	528.4
Case 5 (MIKF)	115	550	52.9	2.1	1762.2
Case 6 (M)	115	490.6	54.4	60	2643.9
Case 6 (I)	160	372.5	78.7	108.8	
Case 6 (K)	115	490.6	54.4	60	
Case 6(F)	160	372.5	78.7	108.8	
Case 7 (M)	100	529	50.8	40.2	2242.7
Case 7 (I)	125	484	59.2	51.8	

Case 7 (K)	50	558.5	50	61.5	
Case 7 (F)	65	573.5	46.2	35.3	
Case 8 (M)	84	426	69.8	140.2	1517.3
Case 8 (I)	133	410.5	54.1	122.4	
Case 8 (K)	173	427.5	68.9	50.6	
Case 8 (F)	202	365	72.6	80.4	
Case 9 (M)	85	497.7	41.1	96.2	1620.5
Case 9 (I)	90	497.9	45.2	86.9	
Case 9 (K)	90	491.7	47.9	90.4	
Case 9 (F)	95	492.9	44.8	87.3	

3.4 Strengths and weaknesses

Strengths:

- 2-D Gantt chart can effectively do scheduling while meeting the preferences of all friends

- Proposed happiness mathematical model can evaluate the scheduled cases enjoyment level

- To increase the enjoyment of the trip, our model can arrange the friend stay together as much time as possible, while meet all preferences.

Weakness:

- Happiness mathematical model coefficients was selected by scheduling experiences, we need calibrate these coefficients with more trip cases.

- Some of the single person charts have too few restrictions thus ending up impractical and unmanageable experiences - such as 10hrs of games. We tried to mitigate this by planning redundant activities

3.5 MidMCM team trip schedule

Based on our proposed methodology in scheduling for the group of friends, we also planned out a day of fun for ourselves.

For our team MidMCM, the planning process will be slightly different as we do not have any constraints such as motion sickness or food allergy. Thus, our planning can be solely based on our preferences, and we also do not have to split up as we have similar preferences. We prefer to make the most out of our time but also include

sufficient buffer times to ensure that we enjoy ourselves to the fullest and not be too packed to a point such that we feel pressured.

We will take a long time to finish our food so we inculcated it into our plan. Since we want to avoid waiting time as much as possible, we will spend our time in the morning doing the extreme thrill rides, when there is no waiting time. Then we will proceed to the medium rides, followed by basic and water rides. After that, we will proceed to the Triangle Restaurant to eat for about 1.5hrs. Straight after eating, we will go onto the basic rides before enjoying the extremes once more. Along the way, we will get some snacks from the hot dog and ice cream stall. Since we prefer rides and are not interested in the Trapezoid Show, we will not spend time watching the shows. After dinner at Triangle Restaurant, we will proceed to the gift shop to end the day.

This is different from the plan for the group of friends as we do not have any food restrictions, for example, everyone of us is fine with eating hotdogs while Karine do not eat hotdogs. None of us has motion sickness, hence, we are all fine with taking extreme thrills and medium thrills after having lunch. None of us really is interested in the Magic and Circus Acrobatics show, hence, we will not watch the Magic and Circus Acrobatics show.

Chapter 4 Conclusion and future development

Conclusion:

Overall, we have accommodated the needs and constraints of the four friends and have made an optimal schedule for each of them. We have accommodated their needs and made a Gantt Chart for each of them. According to our analysis, we can conclude:

- Gantt Chart is an efficient method for scheduling.
- Happiness mathematical model can be applied for the assessment of scheduling.
- We have also compared the schedule we designed for the four friends with the schedule we would have designed for ourselves.
- Even though we have done the job required, there is always space for future development. Such as making the Gantt chart in 3d.

Future Development:

- Apply 3D Gantt Charts for a more complex scheduling [3]
- Apply CPLEX tool for this scheduling problem and find the optimum solution
- Plan multiple cases with different objectives such as walking time, activity time, waiting time while fulfilling all the requirements stated by the question and the group of friends.

Chapter 5 A letter to the group of friends

Dear Karine, Ishmael, Ming, and Freya,

We hope you are looking forward to a day at Polygon Paradise Park. We are writing to share with the optimal schedule we have planned for your group visit to Polygon Paradise Park.

At first you will be split up into two groups of two for a total of 2 hours. Ming will be with Karine for some games and rides ranging from basic to medium thrill. First you will try your hands at some games then the concave train followed by the dodecagon rapids and the starburst carousel. For Ishmael and Freya, you guys will feel the wind in your hair as you take the Decagon coaster twice followed by the Heptagon coaster three times. Then the hexa-swings twice and the pentagon bumpers three times. Followed by Square scramble twice and rhombus ride and concave train twice. As the saying goes "once is never enough."

At 11:00am, you will all reunite to have ice cream for around 15 minutes before going to Octagon Flumes and Dodecagon Rapids. After which, you will be splitting into the same pairs again. Group 2 would go to the Trapezoid show to watch Circus Acrobatics, as Karine likes to see the Circus Acrobatics, and Group 1 would go on the Heptagon Coasters twice, then Decagon Coaster 3 times as Ishmael would like to ride as many coasters as possible. Next, group 1 will ride Rhombus Rides 3 times and Square Scramble 2 times. After both groups have completed the activities, you will merge together and ride concave train.

After that, you will ride Pentagon Bumpers and Hexa-Swings before finally having lunch at the Triangle Restaurant for 50 minutes. After lunch, you will be split again in your respective groups. Group 2 would play games for 50 minutes while Group 1 would go for Dodecagon Rapids and Rhombus Ride before both groups would get back together. As a group, you will be going to the Trapezoid Show to watch the magic show as Ishmael wants to watch them. After the show, you will enjoy Ice Cream before proceeding to ride the Starburst Carousel 3 times before riding the Square Scramble and then Pentagon Bumpers. Next, you will play games for 15 minutes and then proceed to have dinner at Triangle Restaurant for 65 minutes before finally visiting the Gift Shop for 50 minutes, as Freya wants to buy a souvenir before leaving.

This schedule offers the best possible day as it caters the needs of all of you as much as possible. In the schedule, Ishmael and Karine get to watch their Magic show and Circus Acrobatics show and Karine does not eat hotdogs. Ming plays his games and does not get motion sickness and Freya gets her water rides and gets to buy her souvenir. We hope you can enjoy this wonderful trip with your friends. If you have any questions, please feel free to contact us.

Yours faithfully,

MidMCM Travel Agency

Reference

[1] Michel X. Goemans & David P. Williamson "Two-Dimensional Gantt Chart and A Scheduling Algorithm of Lawyer" Siam J. Discrete Math. Vol. 13, No. 3, pp. 281-294

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[3] Christopher V. Jones "The Three-Dimensional Gantt Chart" Operations Research, Nov. - Dec. 1988, Vol. 36, No. 6 (Nov. - Dec., 1988), pp. 891-903